On the Tangent of the Argument: Closed Forms for Constants Arising from Anticommutative Pauli CMF Matrix Products

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Abstract

In a companion paper [1] we introduced anticommutative Conservative Matrix Fields (CMFs) on the Gaussian integer lattice whose step matrices lie in the Pauli algebra, and identified five numerical constants produced by matrix products that could not be expressed in closed form. Here we provide exact closed forms for all five. The key result is a theorem connecting the Pauli CMF recurrence to an alternating arctangent series via a Möbius angle recursion: the constant equals |tan(S)| where S = Σ(−1)^(j+1) arctan(p(j)/r(j)). All five constants share the unified form c = |tan(arg(Γ(1+w/2)/Γ(1/2+w/2)))| = |Im(z)/Re(z)| for a complex parameter w and z = Γ(1+w/2)/Γ(1/2+w/2). We further prove the exact identity $S_1 + S_3 = \pi/4$, which relates two of the five constants. We also establish that S_1 is the phase of a zeta-regularised functional determinant on the circle, and report extensive negative PSLQ searches at the stated precision and coefficient bounds, ruling out relations to Dirichlet L-values, polylogarithms, Bloch-Wigner functions, and Mahler measures.

Keywords: Conservative Matrix Field, Pauli algebra, complex Gamma function, Möbius transformation, arctan series, zeta-regularised determinant, Hurwitz zeta function

1. Introduction

The companion paper [1] extended the CMF framework of Raz et al. [2] to the Gaussian integer lattice $\mathbb{Z}$[i], showing that anticommutative paths admit a natural realisation in the Pauli algebra. Five constants generated by explicit polynomial step matrices resisted identification. The present paper provides exact closed forms for all five, proves an exact relation between two of them, interprets them spectrally, and reports a systematic numerical investigation of their arithmetic nature. Section 2 establishes the Möbius Angle Theorem. Section 3 derives exact Gamma closed forms. Section 4 proves S_1+S_3 = π/4. Section 5 gives the spectral interpretation. Section 6 reports extensive negative PSLQ searches.

2. The Möbius Angle Theorem

2.1. Role of the Pauli structure

The Pauli anticommutation $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$I constrains the CMF step matrices to trace-zero real combinations of two Pauli generators, which in suitable coordinates take the form M = rσ_z + pσ_x = [[r, p], [p, −r]]. Two properties of this form are consequences of the anticommutation, not free parameters:[1]

(i) Involution. $M^2 = (r^2 + p^2)I$, so the induced projective action $z \mapsto (rz + p)/(pz - r)$ satisfies $f \circ f = \mathrm{id}$. This is what produces the alternating sign in S = Σ_j (−1)^(j+1) arctan(p(j)/r(j)) below; generic Möbius iteration does not telescope into sums of arctangents.

(ii) Tangent linearisation. The antidiagonal-plus-diagonal form yields a one-step arctan increment arctan(p/r) when the argument of z is tracked, collapsing two-dimensional matrix dynamics on $\mathbb{CP}^1$ into one-dimensional additive angle dynamics.

The anticommutation thus supplies exactly the algebraic data required for (i) and (ii). Without it, the step matrices would produce generic Möbius iterates whose orbits do not admit closed forms of the type derived here. However, once the arctan sum is obtained, its reduction to a Gamma-function ratio (Lemma 2.2 below) proceeds by classical special-function identities and does not rely further on the Pauli framework. The companion paper [4] correspondingly treats the resulting Gamma-ratio formula as an independent object, with the Pauli origin appearing only as motivation.

Theorem 2.1. The accumulated Pauli CMF matrix ratio z_N satisfies the Möbius recurrence z_k = (r(k)z + p(k))/(p(k)z − r(k)), conjugate to the angle recursion φk = *arctan(p(k)/r(k))* − φ{k−1} via z_k = −tan(φ*k). The CMF constant is c = |tan(S)| where S =* Σ{j≥1} (−1)^(j+1) arctan(p(j)/r(j)).

The proof follows by substitution and the tangent subtraction identity. □

Lemma 2.2 (Gamma Product Formula). For Re(w) < 1:

$$\prod_{k=1}^{N} \left(1 + \frac{w}{k}\right)^{(-1)^{k+1}} \to \sqrt{\pi} \cdot \frac{\Gamma(1 + w/2)}{\Gamma(1/2 + w/2)} \quad \text{[even-}N\text{ limit]}$$

The factor $\sqrt{\pi}$ arises from the Weierstrass product normalisation at the half-integer shift. Since $\sqrt{\pi}$ is real and positive, arg($\prod$) = arg(Γ(1+w/2)/Γ(1/2+w/2)); all downstream uses in Section 3 depend only on the argument and are unaffected.

2.3. Convergence and precision

The alternating arctangent sum S = Σ_j (−1)^(j+1) arctan(p(j)/r(j)) is identified by Lemma 2.2 with arg(Γ(1+w/2)/Γ(1/2+w/2)), the even-N limit of the Weierstrass product. The Leibniz criterion applies directly only to c_1, where p(k)/r(k) = 1/k → 0 monotonically; for c_2, c_3, c_4, c_5 the term arctan(p(k)/r(k)) does not tend to zero (the ratios p/r approach ∞, ∞, 2, and 1 respectively as k → ∞), so the arctangent series is meaningful for these constants only as the formal angle-sum of the Möbius iteration of Theorem 2.1, with convergence supplied by the Gamma-ratio identification. See Table 1, column 3 for the limits at a glance.

For c_1 (p=1, r=k) this bound is 1/(N+1), so direct summation requires O(10^120) terms for 120-digit precision — plainly impractical. In practice, all five constants are evaluated via the closed-form Gamma ratio (Lemma 2.2 and Section 3), using mpmath's internal Spouge approximation for Γ at complex arguments. Spouge's method requires O(120 ln 10) ≈ 277 summation terms for 120-digit relative error, making high-precision evaluation straightforward. The Weierstrass product of Lemma 2.2 converges absolutely since $\Sigma|w/k|^2 = |w|^2\pi^2/6 < \infty$; the product-to-Gamma identification is a standard consequence of the Weierstrass factorisation [7, §12.1].

2.4. Branch conventions

We use the principal branch Arg : $\mathbb{C}^* \to (-\pi, \pi]$ throughout. The constant c = |tan(Arg(z))| where z = Γ(1+w/2)/Γ(1/2+w/2) admits the manifestly branch-free equivalent form c = |Im(z)/Re(z)| (valid when Re(z) $\neq$ 0). Numerical evaluation at 120-digit precision confirms Re(z) > 0 for all five constants (see companion paper [4, §2.3] for the general statement). For these constants Arg(z) $\in (-\pi/2, \pi/2)$, so no branch discontinuities arise in any identity proved below.

3. Exact Gamma Closed Forms

Applying Theorem 2.1 and Lemma 2.2 to each constant yields exact closed forms. The $\sqrt{\pi}$ factor drops from all arguments since it is real positive.

3.1. c_1 [p=1, r=k]: factor 1+i/k gives w = i directly.

$$c_1 = \tan\left(\arg\left(\frac{\Gamma(1+i/2)}{\Gamma(1/2+i/2)}\right)\right) \approx 0.554996111573619503426 40\ldots$$

3.2. c_3 [p=k(k+1), r=k]: factor i(k+(1−i)), i-prefactor vanishes at even N.

$$c_3 = \left|\tan\left(\arg\left(\frac{\Gamma(1-i/2)}{\Gamma(3/2-i/2)}\right)\right)\right| \approx 0.286176849648869555395 22\ldots$$

3.3. c_4 [p=2k−1, r=k]: factor (1+2i)(1−(2+i)/(5k)), even-N removes (1+2i).

$$c_4 = \left|\tan\left(\arg\left(\frac{\Gamma(4/5-i/10)}{\Gamma(3/10-i/10)}\right)\right)\right| \approx 0.247582218708610698892 36\ldots$$

3.4. c_5 [p=k(k+1), r=k^2]: factor (1+i)(1+(1+i)/(2k)), even-N removes (1+i).

$$c_5 = \left|\tan\left(\arg\left(\frac{\Gamma(5/4+i/4)}{\Gamma(3/4+i/4)}\right)\right)\right| \approx 0.207879576350761908546 95\ldots$$

3.5. c_2 [p=$(2k-1)^2$, r=k]: ratio p/r is quadratic. Roots of $4iz^2+(1-4i)z+i=0$ are $r_{1,2} = [-1+4i \pm \sqrt{1-8i}]/(8i)$.

$$c_2 = \left|\tan\left(\arg\left(\sqrt{\pi}\cdot\frac{\Gamma(1-r_1/2)\cdot\Gamma(1-r_2/2)}{\Gamma(1/2-r_1/2)\cdot\Gamma(1/2-r_2/2)}\right)\right)\right| \approx 0.743286453901776199686 60\ldots$$

Table 1. The five constants, their convergence behaviour, and Gamma closed forms.

4. The Identity $S_1 + S_3 = \pi/4$

Numerical verification. At 100-digit precision:

$$S_1 = 0.5066709032166229819852558047835815124728435473470205829 2000\ldots$$

c	$p(k), r(k)$	$\lim p/r$	w	Closed form (arg of)
c_1	$1,\ k$	$\to 0$	i	$\Gamma(1+i/2)/\Gamma(1/2+i/2)$
	value $= 0.5549961115736195034264 0\ldots$			
c_2	$(2k-1)^2,\ k$	$\to \infty$	quadratic	$\sqrt{\pi}\cdot\Gamma$-product
	value $= 0.74328645390177619968660\ldots$			
c_3	$k(k+1),\ k$	$\to \infty$	$-i+$ shift	$\Gamma(1-i/2)/\Gamma(3/2-i/2)$
	value $= 0.28617684964886955539522\ldots$			
c_4	$2k-1,\ k$	$\to 2$	$-(2+i)/5$	$\Gamma(4/5-i/10)/\Gamma(3/10-i/10)$
	value $= 0.24758221870861069889236\ldots$			
c_5	$k(k+1),\ k^2$	$\to 1$	$(1+i)/2$	$\Gamma(5/4+i/4)/\Gamma(3/4+i/4)$
	value $= 0.20787957635076190854695\ldots$			

$$S_3 = 0.278727260180825327630405041036294208576448802496755872323 73\ldots$$

$$S_1 + S_3 = 0.78539816339744830961566084581987572104929234984377645524373\ldots = \pi/4. \quad \checkmark$$

Theorem 4.1. $S_1 + S_3 = \pi/4$, where $S_1 = \arg(\Gamma(1+i/2)/\Gamma(1/2+i/2))$ and $S_3 = \arg(\Gamma(1-i/2)/\Gamma(3/2-i/2))$.

Proof. For the numerator product:

$$\Gamma(1+i/2)\cdot\Gamma(1-i/2) = (1/4)\cdot\Gamma(i/2)\cdot\Gamma(-i/2) = \frac{\pi}{2\sinh(\pi/2)}$$

which is real and positive (reflection formula at $z = i/2$). For the denominator, $\Gamma(3/2-i/2) = (1/2-i/2)\cdot\Gamma(1/2-i/2)$ and the reflection formula at $z = 1/2+i/2$ gives $\Gamma(1/2+i/2)\cdot\Gamma(1/2-i/2) = \pi/\cosh(\pi/2)$, so the denominator product is $(1/2-i/2)\cdot\pi/\cosh(\pi/2)$. The full ratio is:

$$\frac{\pi/(2\sinh(\pi/2))}{(1/2-i/2)\cdot\pi/\cosh(\pi/2)} = \coth(\pi/2)\cdot\frac{1+i}{2}$$

Since $\coth(\pi/2) > 0$, the argument is $\arg(1+i) = \pi/4$. □

Corollary 4.2. $\arctan(c_1) + \arctan(c_3) = \pi/4$, giving $c_3 = (1-c_1)/(1+c_1) \approx 0.28618\ldots$

Remark 4.3. The algebraic relation between c_1 and c_3 reflects the complex-conjugation symmetry of the construction: the step polynomial ratio for c_3 is $j+1$ (a shift of $1/j$ for c_1), and the parameter w changes from i to its conjugate (with a half-integer shift). The $\pi/4$ identity is the angle-space manifestation of this conjugation symmetry.

Remark 4.4. The identity $S_1 + S_3 = \pi/4$ is a special case of a general closed form established in companion Paper 3 [4, Theorem 3.4]: for any $w = \pi/N(\pi)$ with π a Gaussian prime of norm ≥ 2 (and for the unit $w = i$, treated here as the degenerate $N = 1$ case), the combination

$$\arg\left[\frac{\Gamma(1+w/2)\cdot\Gamma(1+\bar{w}/2)}{\Gamma(1/2+w/2)\cdot\Gamma(3/2+\bar{w}/2)}\right] = \arctan\left(\frac{b}{N+a}\right)$$

is an elementary function of (a, b, N). For w = i we have (a, b, N) = (0, 1, 1) and arctan(1/(1+0)) = π/4, recovering Theorem 4.1. The proof in [4] uses only $\Gamma(\bar{z}) = \overline{\Gamma(z)}$ and $\Gamma(3/2+z) = (1/2+z)\Gamma(1/2+z)$, and is algebraically lighter than the hyperbolic-function proof presented here, though the latter makes the geometric origin of the π/4 transparent.

5. Spectral Interpretation

Let D = −id/dx be the momentum operator on the circle S^1. The zeta-regularised determinant det(D+a) = exp(−ζ'(0,a)) satisfies det(D+a) $\propto \Gamma(a)^{-1}$.

Theorem 5.1. S_1 = −arg(det(D+1/2+i/2)/det(D+1+i/2)) = Im[log Γ(1+i/2) − log Γ(1/2+i/2)].

An equivalent form: S_1 = Im[ζ'(0, 1+i/2) − ζ'(0, 1/2+i/2)] via the Lerch formula ζ'(0,a) = log Γ(a) − ½ log(2π). This connects to the arithmetic questions in §7.

6. Negative PSLQ Searches

Extensive PSLQ searches at the stated precision and coefficient bounds found no relations in any of the following classes (mpmath Bailey–Broadhurst pslq, mp.dps = 120, maxcoeff = 10^6; see Appendix A for full details):

6.1. Standard bases {S_1, π, log 2, Catalan, γ, Γ(1/4), Γ(3/4), AGM(1, $1/\sqrt{2}$)}: no relation. S_1/π, S_4/π, S_5/π fail algebraicity to degree 20.

6.2. Im(log Γ(a/N+ib/M)) for N $\leq$ 9, M $\in$ {2,4,5,6,8,10}: 273 values, no relation.

6.3. Im(Li_n(z)) for n = 2–6 and algebraic z: no relation.

6.4. Bloch-Wigner D_2(z) and Zagier's D_3(z): no relation.

6.5. Complex Mahler measures of 1+x+y+αxy: no relation.

6.6. Dirichlet L-values L(1,χ) and L(2,χ) for conductor $\leq$ 12: no relation.

6.7. Pairwise ratios S_4/S_1, S_5/S_1, S_5/S_4 not algebraic to degree 15.

These null results are evidence, not proof, of non-existence of low-height algebraic relations; see Appendix A for the height-bound context.

7. Open Questions

(1) Shimura/Deligne framework. Is Γ(1+i/2)/Γ(1/2+i/2) a period ratio of a CM abelian variety associated to $\mathbb{Q}$(i)?

(2) Generalised identity. Does $S_4 + S_5$ satisfy an analogue of Theorem 4.1?

(3) Hurwitz zeta derivatives at complex arguments. Does ζ'(0,a) for complex a have arithmetic significance analogous to the rational case?

(4) Infinite family. Any rational p(k)/r(k) produces a constant |tan(arg(Γ(1+w/2)/Γ(1/2+w/2)))| for some w $\in \mathbb{Q}$(i). Classifying this family is a natural programme — see companion Paper 3 [4].

8. Conclusion

All five constants have exact closed forms $c = |\mathrm{Im}(z)/\mathrm{Re}(z)|$ where $z = \Gamma(1+w/2)/\Gamma(1/2+w/2)$. Two satisfy $S_1 + S_3 = \pi/4$ (proved). S_1 has a spectral interpretation as the phase of a zeta-regularised determinant. Extensive PSLQ searches at 120-digit precision across seven classes of known constants found no algebraic relations. Each constant is verified absent from the OEIS, Wolfram Alpha, and ISC databases at the precision tested.

Python/mpmath verification (all five closed forms, 50-digit precision):

[] import mpmath; mpmath.mp.dps = 50 def c(w): z = mpmath.gamma(1+w/2)/mpmath.gamma(0.5+w/2) return abs(mpmath.im(z)/mpmath.re(z)) # branch-free form c1 = c(mpmath.mpc(0,1)); c5 = c(mpmath.mpc(0.5,0.5)) S1 = mpmath.arg(mpmath.gamma(1+1j/2)/mpmath.gamma(0.5+1j/2)) S3 = mpmath.arg(mpmath.gamma(1-1j/2)/mpmath.gamma(1.5-1j/2)) print(S1 + S3) # prints pi/4 to full precision

Remark on Clifford generalisation. The collapse mechanism identified above — anticommutation forces a trace-zero matrix form whose projective action is involutive (§2.1), producing alternating arctan sums — is specific to Cl(2). Whether an analogous collapse exists for higher Clifford algebras Cl(p, q) acting on spinor spaces depends on identifying an observable whose dynamics reduce to a tractable scalar form. We do not address this question here.

Appendix A: PSLQ Computational Details

All PSLQ searches used the mpmath Bailey–Broadhurst implementation (mpmath v1.3+, mp.dps = 120 with 20-digit guard, maxcoeff = 10^6). The basis sets in §6.1–6.7 were chosen as arithmetically natural: closed under Galois conjugation over $\mathbb{Q}$ and spanning the known Class I–II gamma values at denominators ≤ 12. A null result at maxcoeff 10^6 rules out integer relations with coefficient height below 10^6; the Ferguson–Forcade theory [W. Ferguson, R. Forcade, *Math. Comp.* 40 (1983)] implies the actual relation height, if one exists, exceeds 10^6. Full scripts, target values at 120-digit precision, and configuration files are available at github.com/Loublain/cmf-gaussian-primes [archived: doi:10.5281/zenodo.19670531].

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Footnotes

[1] The Pauli anticommutation relations and the anticommutative CMF condition S + T = 0 are formalized in `SuperCMF.lean` and `PauliCMF.lean` at github.com/Loublain/cmf-gaussian-primes. The algebraic sign flip mechanism is proved as `AntiCMF.sign_flip`.